PL05 · PLAY LAB TRACK

IDEATION PHASE

LEARN YOUR WAY OUT

Every learning game is a quiz in a costume.
*In ours, **the maze is the quiz.***

TEAM	PROJECT	TRACK
Solo	PL05 – Learn Your Way Out	Play Lab

- 3D browser game
- AI custom questions
- Live multiplayer
- Adaptive difficulty



EARLY CONCEPT PROTOTYPE

The maze is the quiz

Most learning games pause the fun for a quiz popup. Here, every answer is a place you run to.

Answer with your feet
5 room types turn every question type into movement.

Wrong turns teach
A detour shows the answer, why, a worked example, a retry.

Adapts to every player
Harder after 2 right, easier after a miss; replay weak spots.

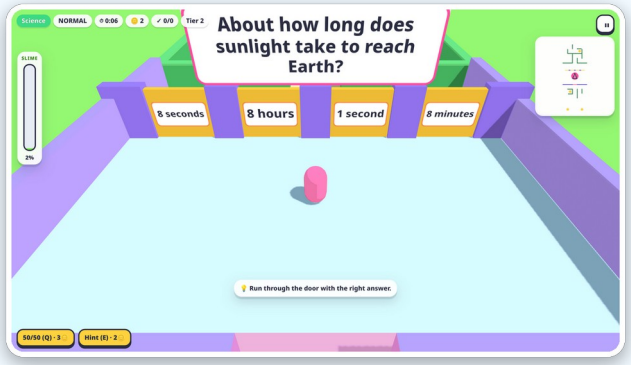
AI custom questions
Type a topic or paste notes; DeepSeek builds a live pack.

Live multiplayer races
Race friends through the same seeded maze as ghost beans.

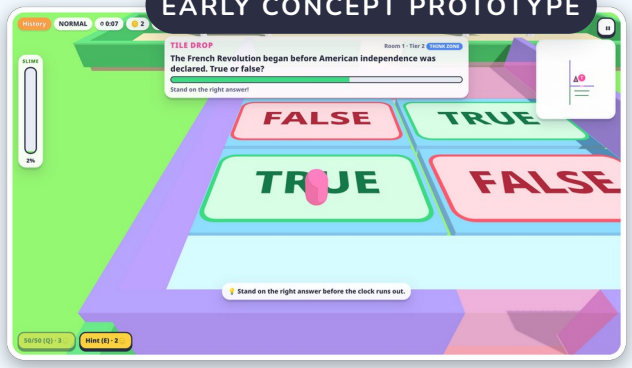
A new maze every run
Seeded mazes with forks, dead ends, coins and rising slime.

USP Answers are places, not buttons. Mistakes become lessons, not game-overs.

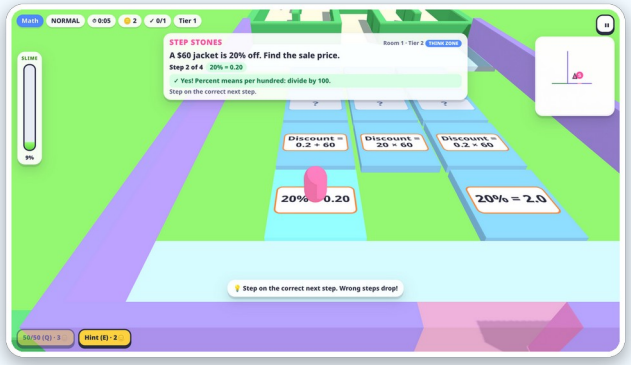
EARLY CONCEPT PROTOTYPE



DOOR RUSH multiple choice



TILE DROP true / false



STEP STONES multi-step problems



BRIDGE BUILDER put in order

One loop, from pick to escape



PLANNED BROWSER CLIENT

- **UI** React DOM + zustand: HUD, menus, results
- **Render** three.js + React Three Fiber
- **Engine** TypeScript, fixed 120 Hz step, own collision
- **Content** JSON packs + validator, QuestionSource

AI QUESTIONS

Node proxy → DeepSeek API, key stays on server

MULTIPLAYER

WebSocket relay: shared seed + positions 10/s

- **Maze:** DFS backtracker + braiding on a seeded PRNG: one seed, one identical maze.
- **Adaptive picker:** tier staircase, scored by recency and weak concepts.
- **AI pipeline:** JSON schema → validator → cached pack; built-in bank as fallback.
- **Multiplayer:** share the seed, stream positions; each client runs its own course.
- **Test plan:** a headless bot plays every generated course before release.

Feasible, low-cost, low-risk

4
domains at launch: math, science, history, programming

5
room types, one for each question format

3
difficulty levels, plus adaptive tiers

0
installs: runs on laptops and phones from a link

ROADMAP

- **Concept & design** — WE ARE HERE
  brief, mechanics, architecture
- **Core prototype**
  maze, 5 rooms, detour, slime
- **Content & adaptivity**
  question packs, tiers, weak spots
- **AI custom questions**
  DeepSeek packs from topics or notes
- **Multiplayer & pilot**
  live races, a classroom trial

FEASIBILITY

- Proven web stack: three.js + React
- Pure browser, touch for phones
- Modular: AI + multiplayer plug in

VIABILITY

- Core is static files: free hosting
- One AI call = one cached pack
- Relay forwards positions only

RISKS → MITIGATIONS

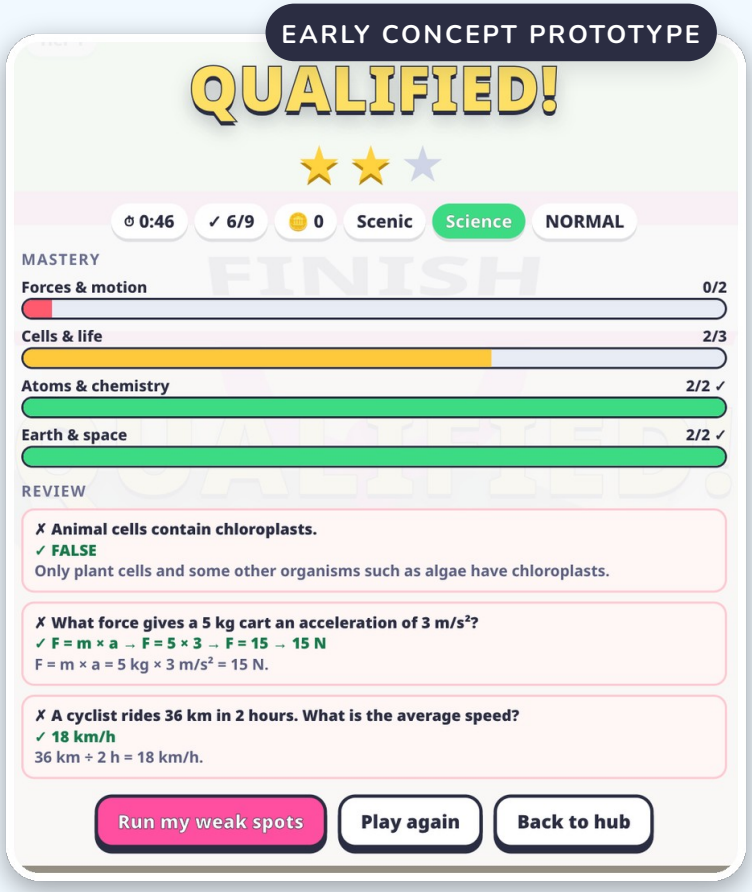
Wrong AI facts
schema validator, second-pass check, teacher review

AI latency
generate in the hub before the run; built-in bank fallback

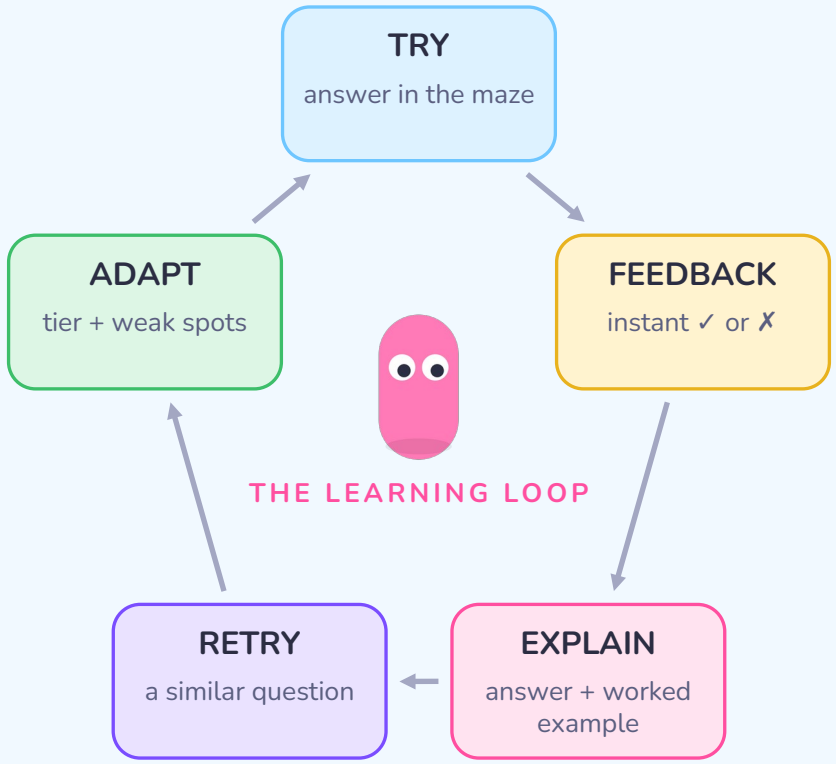
Leaked API key
key stays on a server proxy, never in the browser

Lag in races
each client simulates locally; ghosts interpolate

Mistakes become lessons



Planned: every run ends with mastery per concept and a review of every miss.



Students

Active recall in every room, safe low-stakes mistakes, and races with friends.

Teachers

Questions from your own notes in seconds, plus a weak-spot report per concept.

Schools

No installs, runs on old laptops and phones; read-aloud, reduced motion, touch.

Grounded in learning science

RESEARCH SAYS	SO THE GAME DOES
Testing effect *Roediger & Karpicke, 2006*	Every room is a retrieval attempt, answered by moving
Feedback *Hattie & Timperley, 2007*	Instant ✓ or ✗ on the door, tile or step you chose
Learning from errors *Metcalfe, 2017*	A wrong answer opens a detour: answer, why, worked example, retry
Desirable difficulty & flow *Bjork & Bjork, 2011; Csikszentmihalyi, 1990*	The adaptive tier keeps the challenge just above skill
Spacing *Cepeda et al., 2006*	Weak-spot runs bring missed concepts back later
Games that teach *Wouters et al., 2013; Clark et al., 2016*	The learning is the core mechanic, not a popup

REFERENCES

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5 Csikszentmihalyi (1990). Flow: The Psychology of Optimal Experience.

6 Cepeda et al. (2006). Distributed practice in verbal recall tasks. Psychological Bulletin, 132(3).

7 Wouters et al. (2013). Cognitive and motivational effects of serious games. J. Educational Psychology, 105(2).

8 Clark, Tanner-Smith & Killingsworth (2016). Digital games, design, and learning. Review of Educational Research, 86(1).

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11 Docs: three.js, React Three Fiber, DeepSeek API. Art direction: Fall Guys (Mediatonic, 2020).